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Palaeomagnetism of Tertiary Basalts from Libya

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Summary

Of the Tertiary volcanics in Jebel Nefousa (Garian area) and Jebel Soda, 42 individual lava flows and outcrops were sampled providing a total number of 457 samples for palaeomagnetic measurements. With the exception of three phonolites in the Garian area, the volcanics can all be classified as alkali basalts.

After selective AF-cleaning in fields between 110 and 500 Oe, 20 sites (218 samples) from the Garian area were used for calculation of a palaeomagnetic pole at 86·1 °N, 152 °E with a circle of confidence $A_{95} = 3{\cdot}7°$. From the Jebel Soda area, 12 sites (138 samples) were used to define a palaeomagnetic pole at 78·4 °N, 196 °E with $A_{95} = 7{\cdot}4°$. The poles from both areas are different from the present geographic pole and are in general agreement with other results from African rocks during Miocene to Eocene.

1. Introduction

In the western and south-western part of the Sirte Basin in central Libya there are three extended areas of mostly Tertiary volcanism: Jebel Nefousa (Garian area), Jebel Soda and Jebel Harudj (Fig. 1). The palaeomagnetic studies presented here are confined to samples from the Garian and Jebel Soda area. With the exception of some dome-like phonolite bodies near Garian, all the lava flows are of olivine-rich alkali basalts (Klitzsch 1970). The few radiometric age determinations (K–Ar ages) by Piccoli (1971) and unpublished data from oil companies date the volcanism in the Garian area back to the Eocene. However in the more southern part of the Garian area Miocene and Pliocene ages have been determined. As we have used for our final conclusion only samples from sites in the nothern part of that area we assumed an average age of 42 ± 10 My, which is the mean of five K–Ar ages (29, 37, 38, 52 and 53·5 My). Three radiometric age determinations of the phonolites near Garian according to Piccoli (1971) give an average value of 39·4 My (37·9, 39·7 and 40·7 My). The basalts of the Jebel Soda area are in most cases of Miocene age (Klitzsch 1970) but some of Pliocene age may also occur (Cornelius, private communication). The basalts of Jebel Harudj are of Pliocene and younger age (Klitzsch 1970). The ages of the volcanics of Jebel Soda and Jebel Harudj were mainly determined by stratigraphic methods.

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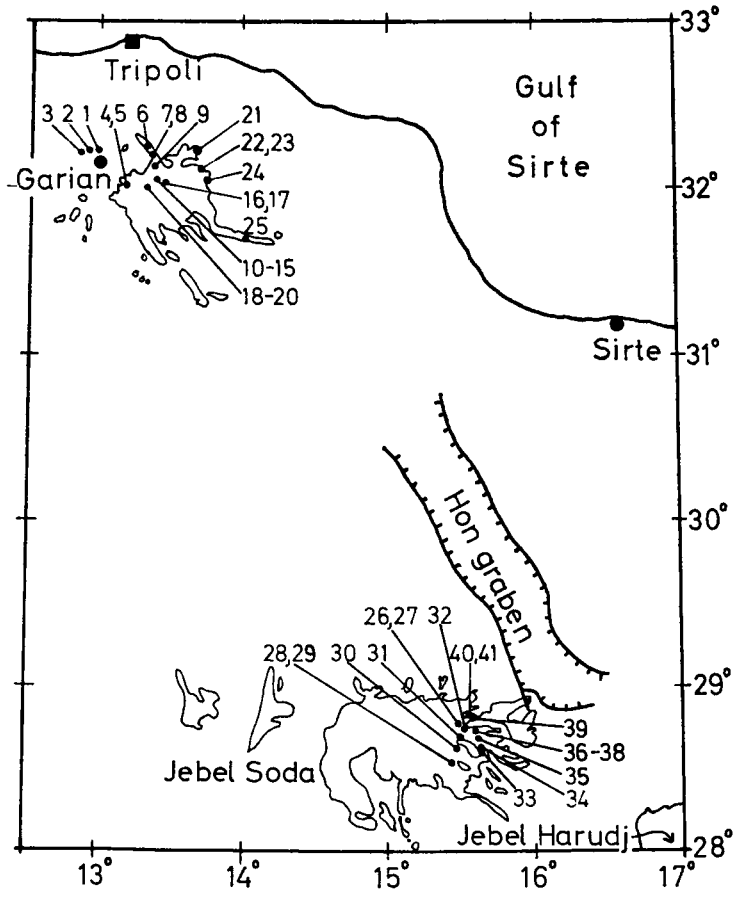


FIG. 1. Map showing location of basaltic volcanism and of sampling sites. Numbers refer to site numbers.

2. Sampling areas

Samples were taken with portable core drilling machines. At each site at least 10 cores about 10 cm long were drilled.

(a) *Garian group*

Twenty-six different lava flows or outcrops were sampled, three of them are phonolites near Garian, the rest are alkali basalts from the region east of Garian. Sampling was done along roads, tracks or wadis.

(b) *Jebel Soda group*

Sixteen lava flows were sampled in that region, all of them alkali basalts. All localities are situated in the east of Jebel Soda which is the most easily accessible part. Sampling was done along roads, tracks and wadis.

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3. Palaeomagnetic results

Natural remanent magnetization

The remanent magnetization was measured with a PAR spinner magnetometer and some of the strongly magnetized samples (NRM larger than 200 . 10^{-5} Gauss) were measured with a flux gate magnetometer. NRM from both areas showed indications of a very stable remanent magnetization because the within-site precision parameters of NRM of samples with reversed polarity were large. The phonolites from the Garian area exhibited very large scatter of NRM directions indicating unstable remanent magnetization components. All samples were subjected to AF-cleaning with fields between 110 and 500 Oe according to results from AF-demagnetization of test samples.

(a) *Results from the Garian area*

The results after AF-cleaning of the volcanics from the Garian area are shown in Table 1. Sites 1, 2 and 3 are phonolites which had to be omitted because of unstable NRM, the scatter of which could not be adequately reduced by AF-cleaning. Of the remaining sites, 23 was discarded because of extremely large within-site scatter compared with the other sites; site 22 was omitted because of its intermediate direction and site 25 because of its too young age (3·5–5·7 My according to Piccoli (1971)). Of the remaining 20 sites 11 show reversed and nine normal polarity. The mean directions and pole positions of normal and reversed and both polarities together are listed in Table 1. Fig. 2 shows the mean pole positions of the individual sites on an equal area stereographic net together with their mean pole positions.

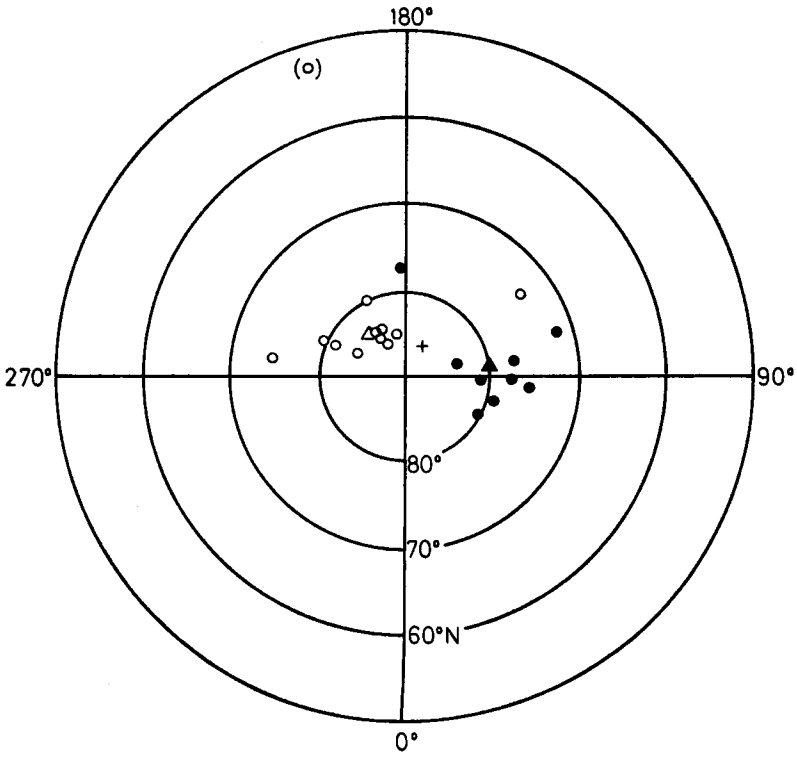


FIG. 2. Equal area stereographic projection showing pole positions of basaltic rocks from the Garian area. ●, ○, site means; ▲, △, mean of all sites with the same polarity. Open symbols refer to reversed polarity, closed symbols refer to normal polarity. +, mean of all sites, irrespective of their polarity. Pole positions in brackets were omitted (see text).

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Table 1

Site mean palaeomagnetic results after AF-cleaning for the Garian basalts (Mean site location is at 32·0° N, 13·4° E)

Site	N	AF(Oe)	RM (in 10^{-5} Gauss)	$D°$	$I°$	$\alpha_{95}(°)$	k	Pole position Lat. °N	Long. °E
1	17	400	1·2	(200·6	−64·3)	15·5	4·8		
2	4	400	0·8	(102·3	−51·3)	23·8	8·7		
3	9	400	0·3	(173·0	−31·5)	23·4	4·0		
4	10	500	150·0	173·9	−41·1	2·7	266·3	80·0	228
5	10	110	160·0	164·4	−43·3	2·8	248·5	74·8	261
9/1	6	110	151·0	170·8	−43·4	3·2	324·8	79·5	246
9/7	8	270	470·0	178·1	−46·0	3·1	249·8	85·1	214
10	11	270	310·0	178·5	−45·0	2·7	237·0	84·4	207
11	11	110	1150·0	180·3	−44·9	2·8	229·2	84·5	191
12	10	330	390·0	174·8	−46·6	3·6	155·5	83·9	242
13	9	110	1010·0	178·7	−46·8	2·2	470·0	85·9	209
16	10	270	290·0	172·4	−44·8	2·9	232·9	81·3	245
20	10	110	270·0	177·2	−44·5	2·4	347·2	83·7	217
21	15	270	120·0	195·9	−42·7	6·4	32·4	74·2	127
6	9	110	240·0	14·5	+53·5	3·5	173·4	77·7	90
7	5	110	810·0	14·7	+51·4	3·6	295·5	77·6	99
8	14	300	260·0	9·2	+57·5	4·4	72·4	80·3	61
14	10	270	200·0	20·8	+48·2	2·7	271·2	72·0	107
15	18	300	220·0	7·2	+50·9	1·9	304·0	83·9	105
17	11	270	400·0	9·9	+53·3	3·2	169·9	81·5	88
18	10	110	400·0	16·4	+54·9	3·0	216·8	76·0	85
19	11	330	170·0	11·3	+56·3	1·9	550·0	79·5	73
24	20	270	350·0	2·2	+35·7	3·6	75·2	77·6	184
22	7	110	990·0	(177·1	+10·7)	8·1	43·0	52·5	198
23	10	110	880·0	(164·2	−54·2)	29·7	2·2		
25	10	110	310·0	(171·4	−34·2)	5·1	75·0		
Mean values:									
All reversed: 11 sites				176·8	−44·7	3·1	190·9	83·7	220·4
All normal: 9 sites				11·5	+51·5	4·3	119·1	80·2	99·6
All sites: 20 sites				182·9	−48·0	3·4	84·2	86·1	152·4

Sites with directions of remanent magnetization in brackets were omitted (see text). N: Number of samples; AF(Oe): peak value of alternating field cleaning; RM: remanent magnetization after alternating field cleaning; D: declination from North to East; I: inclination; α_{95}: radius of the 95 per cent circle of confidence; k: precision parameter.

The mean pole positions of sites with normal and reversed polarity clearly differ from each other and their respective circles of confidence do not overlap. However, all sites (normal and reversed sites) have been considered in subsequent discussion. As the distribution of normal and reversed directions is non-Fisherian, the meaning of the given values of the circle of confidence α_{95} and the precision parameter k, should be used cautiously.

(b) *Results from the Jebel Soda area*

The results after AF-cleaning are listed in Table 2 and Fig. 3. Sites 27, 28, 29 and 31 were omitted because of intermediate directions (latitude of the pole positions less than 50 °N). Of the remaining 12 sites 10 show reversed and two normal polarity. No discrimination between the pole positions of the two groups was made.

The remarkable small within-site scatter (equal weight to each sample) could, in many cases, only be slightly reduced by AF-cleaning, and is characteristic for most of the basalts from both sampling areas, indicating the presence of a very stable remanence.

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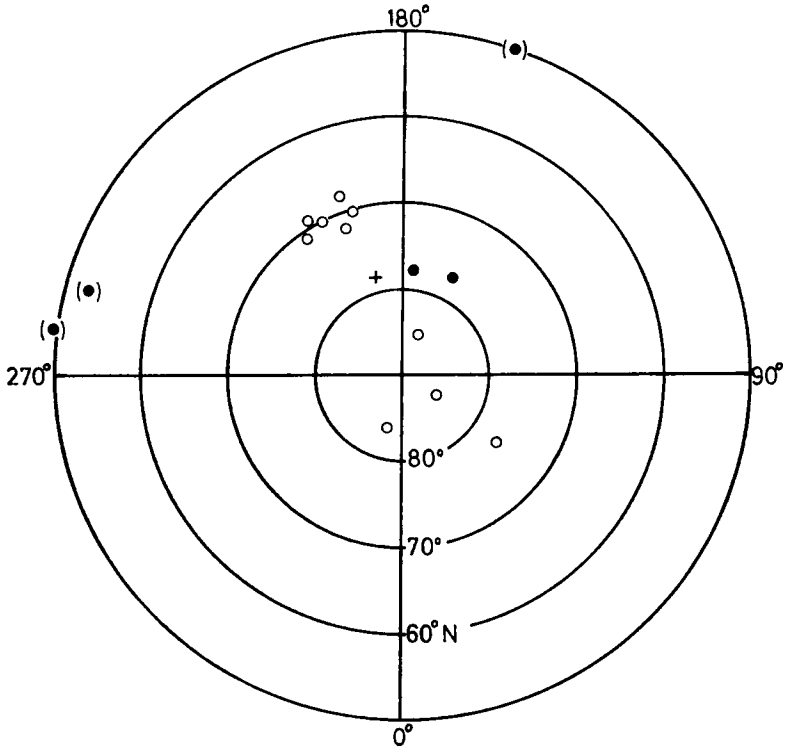


FIG. 3. Pole positions of basaltic rocks from Jebel Soda area. ●, ○, site means; Open symbols refer to reversed polarity, closed symbols refer to normal polarity. +, mean of all sites irrespective of their polarity. Pole positions in brackets were omitted (see text).

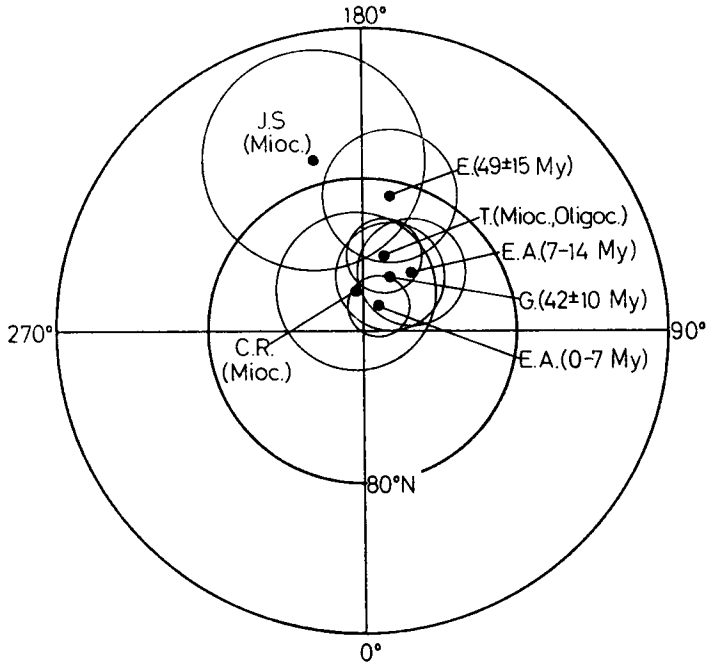


FIG. 4. A selection of Tertiary palaeopole positions with their respective circles of confidence. For references and other details see Table 3. J.S.: Jebel Soda; E.A.: East African Rift Valley lavas; C.R.: Central Rift; T.: Turkana; G.: Garian; E.: Ethiopian flood basalt.

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Table 2

Site mean palaeomagnetic results after alternating field cleaning for the Jebel Soda basalts

(*Mean site location is at* 28·7° *N and* 15·6° *E*)

Site	N	AF(Oe)	RM (in 10^{-5} Gauss)	$D°$	$I°$	$\alpha_{95}(°)$	k	Pole position Lat. °N	Pole position Long. °E
26	12	270	210	183·6	−51·2	3·5	134·0	85·6	59
27	7	250	80	(146·1	+32·4)	4·3	154·0	33·3	235
28	10	250	180	(328·2	+12·0)	2·1	430	52·4	255
29	17	270	70	(323·3	+15·5)	4·0	70·8	49·7	262
30	10	270	150	174·5	−16·4	3·5	159	69·8	211
31	10	270	100	(21·1	−11·0)	6·1	52	50·2	162
32	10	110	440	175·8	−17·4	2·0	497	69·8	208
33	10	110	1040	4·6	+32·0	1·9	528	77·9	174
34	11	270	210	8·6	+35·1	3·6	137	77·8	154
35	11	270	490	178·5	−13·4	1·9	508	68·1	200
36	10	110	760	179·2	−17·5	1·8	629	70·2	198
37	10	110	170	173·5	−20·9	3·0	218	71·1	216
38	17	270	310	178·3	−20·7	3·7	83·8	72·0	201
39	14	270	60	183·0	−42·6	5·2	52·8	85·2	161
40	13	270	180	176·0	−53·4	5·0	60	83·8	344
41	10	270	150	190·1	−58·0	2·5	310	78·0	53
Mean of 12 sites:				179·9	−31·6	8·3	23·5	78·4	196·1

Sites with directions of remanent magnetization in brackets were omitted (see text).

Legend see Table 1.

4. Conclusion and discussion

In Table 3 and Fig. 4(a) selection of Tertiary palaeomagnetic data from Africa is compiled. The results obtained from the Garian area as well as from the Jebel Soda area are in agreement with the data of others ranging from Miocene to Eocene age. The circles of confidence overlap and do not include the present geographic pole indicating a significant polar difference in the order of 5°. However, a non-Fisherian distribution of directions or pole positions, a non-axial or a non-geocentric dipole could lead to this result. Wilson (1970) pointed out that palaeomagnetic results from all over the world indicate a persistent off centre displacement of the dipole of about 190 km along the rotational axis over the past two million years and a similar displacement during all of upper Tertiary time. Such a displacement can be the reason for the polar difference when calculating the pole positions on the axial geocentric dipole model. Even if this problem can be solved it seems difficult to derive a polar wander path for Africa during Tertiary from the presently available data because all circles of confidence of the results from rocks of Miocene to Eocene age overlap. Going back in time a polar shift of roughly a few degrees along the 160° meridian is possible but much more extensive palaeomagnetic studies on radiometrically dated rocks are necessary to get a significant result.

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Table 3

Selection of palaeomagnetic data from Africa during the Tertiary (all basaltic rocks)

Locality	Age (My)	N	$\alpha_{95}(°)$	k	$A°$	Pole position Lat. °N	Pole position Long. °E	Reference
Jebel Soda 28·7° N 15·6° E	Miocene, Pliocene	12	8·3	23·5	7·4	79·4	196·1	Present paper
Rift Valley lavas 1·0° S 36·0° E	0–7	120	2·8	22·0	2·1	88·0	149·7	Reilly (1970)
	7·0–14	43			3·5	85·0	140·0	Reilly (1970)
Central Rift, Kenia 0° N 36·5° E	Miocene	24			5·2	87·4	188·0	Raja (1968)
Turkana 3·6° N 35·1° E	Oligocene, Miocene	62	3·1	34·9	2·3	84·8	163·0	Raja (1968)
Garian 32·0° N 13·4° E	42±10	20	3·4	84·2	3·7	86·1	152·0	Present paper
Ethiopian flood basalt 9·0° N 38·5° E	49±15	20	5·6	34·9	4·3	80·8	168·0	Brock *et al.* (1970)

N: Number of sites; α_{95}: radius of the 95 per cent circle of confidence; k: precision parameter; A°: radius of the 95 per cent of confidence for the pole position.

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